DRÄXLMAIER

Interface Description

DRÄXLMAIER Group MES

OPC UA Template



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2 Version History

Version	Date	Modifications	Editor
2.00	12.12.2018	Initial version 2.00	Andreas Straßer
2.01	25.02.2019	Added new charts for standard workflow	Andreas Straßer
2.02	02.05.2019	Reworked overlay and extended some descriptions; changed error to alert in global data	Andreas Straßer
2.03	01.07.2019	Adopted Figure 2: OPC UA Variable Tree to real situation in variable tree	Johanna Graf
2.04	10.07.2019	Added more description for Alert handling	Andreas Straßer
2.05	24.07.2019	Updated Figure 2: OPC UA Variable Tree	Andreas Straßer
2.06	05.11.2019	Added flow charts for auto scan before process start (5.2.2) and inside process (5.2.3); added MES_Scan, ScanFinished and ScannedBarcode as optional fields to variable tree	Johanna Graf
2.07	17.12.2019	Added FileName as optional field to variable tree in ProcessResult section; add an overview of the auto scan inside process workflow	Johanna Graf
2.08	05.02.2020	Extended some descriptions	Andreas Straßer
2.09	18.03.2020	Adopted Figure “Standard workflow with automatic scanner inside process” to scan either before or after MES_Release; added additional description to various variables in OPCUA variable tree	Johanna Graf
2.10	24.06.2020	Added chapter “OPCUA in the DRÄXLMAIER Group”	Andreas Straßer
2.11	10.03.2021	Modifications in the parameter structure - Added Metric section in GlobalData - Removed Counters section - Added column for units - Added ProcessTimes and MaterialConsumption within ProcessResults section	Andreas Straßer
2.12	19.08.2021	Modifications in the parameter structure - Changed structure of MaterialComsumption section - added TotalProcessingTime within GeneralData	Johanna Graf
2.13	01.10.2021	Modifications in the parameter structure - Added energy related parameters to Tracing structure - Extended description of field “FileName” - Changed description of MaterialConsumption structures	Andreas Straßer
3.00	29.03.2022	Updated the description to match the parameter structure. Updated OPC UA Variable Tree Modifications in the parameter structure - Prepared interface description for SAP readyness - Added “EmptyCycle” as optional to ControlParameter section - MaterialConsumption: o Extended field descriptions for SAP o Added parameters SinglepartLabelSupplier & PositionIndex - Removed old Energy related variables (e.g. PowerCons, AirCons) - Renamed MachineNumber -> EquipmentMainNumber - Renamed EquipmentNumber ->	Johanna Graf, Andreas Straßer

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		EquipmentExemplarNumber - Renamed FunctionalGroup -> FunctionalGroups Updated Workflow drawings Added chapter 6 – MES input parameters Added chapter 7 – Process results & return values	
3.01	16.05.2022	Extended description for MaterialConsumption	Andreas Straßer
3.02	20.10.2022	added optional section SecurityData; adopted sorting of fields to same sorting as in nodeset; added more detailed infos fordatatypes “Int” and DateTime (e.g. Int16, LDT,…)	Johanna Graf
3.03	30.05.2023	- Added the OPC UA foundation member logo - Extended description for: o Default values (4.1) o Array sizes in nodeset and interfaces (4.1) o AlertActive (4.5.1) o Tracing data (4.5) - Added chapters: o “3.6 NodeSet.xml files” o “4.3.9 Part counters” - Changed OPC Structure o Changed/added several predefined StringLengths o Renamed Real to Float o Added part counters as described in new chapter “4.3.9 Part counters” o Added description to ActiveEnergy & ActivePower	Andreas Straßer, Johanna Graf

Table 1: Version History

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3 Introduction

3.1 OPC UA in the DRÄXLMAIER Group

OPCUA is set as standard for all machine interfaces in the shopfloor. If new machines are going to be built up they should implement this template. Otherwise new machines cannot be integrated into visualizations or certain reporting tools.
In case the supplier has a predefined standard interface, or there is a fully developed interface using other technologies the technical IT department of DRÄXLMAIER needs to be involved to decide which interface is going to be used.

3.2 OPC UA in general

The abbreviation OPC UA stands for "Open Platform Communications Unified Architecture" and describes a collection of specifications that standardize communication in the field of industrial automation and M2M communication. Both the transport of data and interfaces, security mechanisms and the semantic structure of the data are specified. OPC UA is developed by the OPC Foundation (https://opcfoundation.org/).
Servers used for interfaces in the DRÄXLMAIER Group need to be certified by the OPC Foundation.

3.3 Network separation

The machine internal network needs to be galvanically isolated from the DRÄXLMAIER production network. There are different options to ensure this separation. For example using an embedded OPC UA server which provides this functionality (see picture).

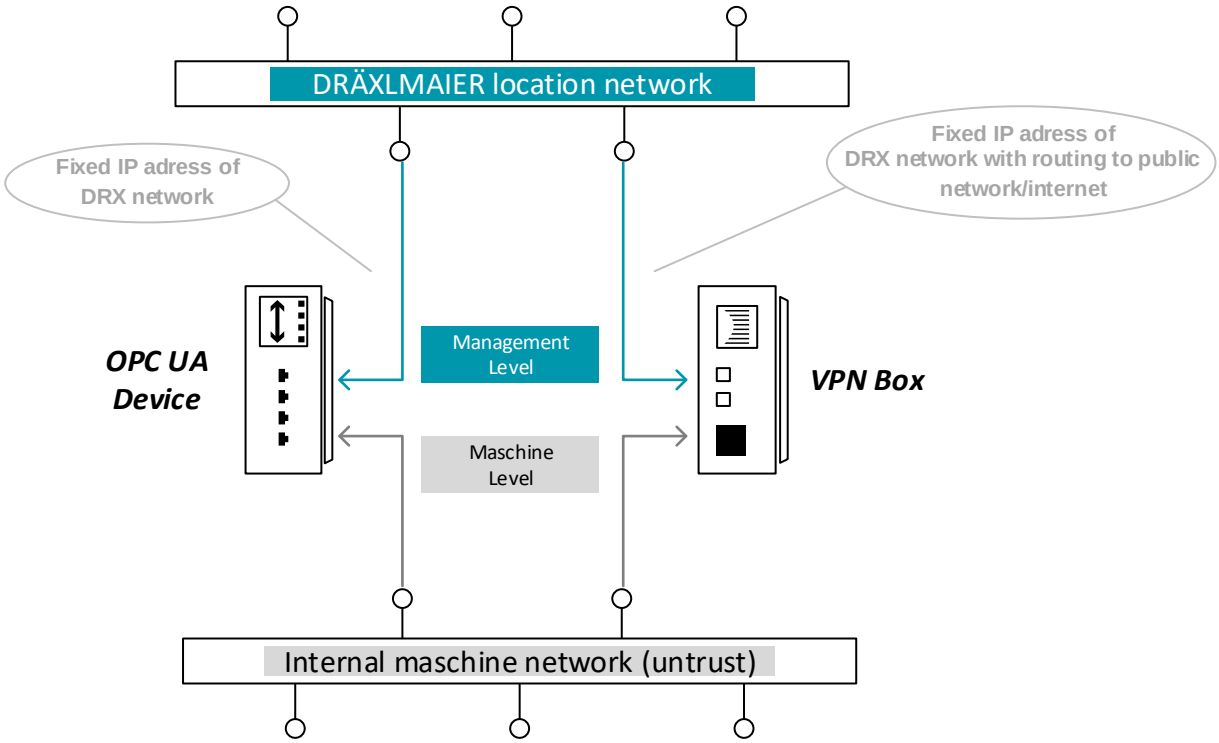


Figure 1: Network setup

Remote access for maintenance reasons can be implemented by installing a VPN Box.
The IP addresses for OPCUA connection and VPN box towards DRÄXMAIER network will be provided by the plant IT. IP addresses used inside the machine network can be freely defined by the machine manufacturer.

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3.4 Certificate handling and security

Currently we do not use the safety features provided by OPC UA, safety is ensured by our network and firewall configurations. Therefore all safety modes shall be accepted by the server (e.g. Authentication: sign, sign & encrypt, none).
The server certificate shall be self-signed and valid for at least 5 years after integration of the machine.

3.5 Hardware configuration

Location of variables is defined by the machine manufacturer. Best practice is to have one communication DB which is mirrored in the OPC-UA server. The structure of the communication DB can be provided as NodeSet.xml

The OPC-UA server itself has to be provided and configured by the machine manufacturer (first setup). OPC-UA addresses of the required variables have to be provided as soon as they are known.
If a Siemens PLC and the internal OPC Server is used here Siemens provides a detailed documentation: https://support.industry.siemens.com/cs/ww/de/view/59192925

Administration of connection, safety features and service options must be accessible from DRÄXLMAIER side. This can be done by providing web masks to access this settings, in this case the login credentials need to be communicated to DRÄXLMAIER. When embedded OPC UA servers of Softing or IBH are built-in their default credentials can be used.

3.6 NodeSet.xml files

Additionally to the Excel description of the interface, DRÄXLMAIER IT can provide the parameter structure of the interface as Nodeset.xml-file.
Nodeset Files are information model schemas, standardized by the OPC foundation. The usage of these files greatly simplifies the exchange of information models and the configuration of clients and servers. The structure no longer has to be created manually, but can be imported into the OPC UA server, which reduces the time as well as the risk of errors in the server creation process.

More information can be found on following website: Using Nodeset Files to Exchange Information – OPC Connect (opcfoundation.org)

4 Communication guidelines

4.1 General topics

- **The communication flow is controlled by various control variables.**
  These variables are organized in an “OPC UA Variable Tree” (see chapter “OPC UA Variable Tree” for more information).

- **DRÄXLMAIER MES acts as a client to the OPC UA server embedded in the assembly line.**
  A Panel PC is installed at the assembly line by DRÄXLMAIER. This Panel PC is connected to the DRÄXLMAIER MES and to the assembly line. CAAFrontend (a software of DRÄXLMAIER) is installed on this PC. This software provides the ability to control the needed interactions between DRÄXLMAIER MES and the assembly line or station.

- **Process data will be exchanged via predefined data exchange variables.**
  The process data of each produced part will be stored in section ProcessResults in the “OPC UA Variable Tree”.

  See chapter “OPC UA Variable Tree” for more information.

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- **Variables with Prefix 'MES_' should only be written by MES, the machine only reads these values.**
  The Prefix highlights that MES is responsible for the content of this variable. The machine should not write or clear these variables in any case.

- **Inside variable and data block names only alphanumerical letters and underline (‘_’) are allowed**
  All other special characters (e.g. german umlauts, $, “, +, -, …) and blanks (‘ ‘) are prohibited!

- **If floating point numeric values are required inside strings, the decimal separator is a ‘.’**
  Example: 3.14159

- **Default values for variables which cannot be filled with an correct value, should be filled as follows:**

OPC data type	Default value
Boolean	false
Numeric	-1
String	Empty string (“”)
DateTime	Systemdefaultvalue („1970-01-01-00:00:00“), or the respective default minimal value of the used data type

- **Array sizes**
  All arrays defined in the nodeset are created with element size 0 by default. Size 0 means that element size can be dynamic from nodeset point of view.
  The real array sizes which have to be used on e.g. the PLC need to be defined during project phase in cooperation with manufacturer and machine ordering persons. The result will be documented in an additional document.
  In the nodeset the element size will stay on 0 to allow future changes on PLC side without the need of a nodeset update.

- **MES_SyncDateTime is a time stamp which is updated once a second by the MES client connected to a station.**
  It can be used by the machine to update the internal time and also as live bit. If this time is not updated for some time, most likely the connection to the client failed or the client crashed.

- **The responsibility of writing OPC UA variables is defined in column “Write Access” in “OPC UA Variable Tree”.**

- **The root node of the OPC UA Tree is defined by the assembly line manufacturer.**
  Best practice is to have one communication DB which is mirrored in the OPC UA server.

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4.2 OPC UA Variable Tree

The OPC UA variable tree is divided into various sections:

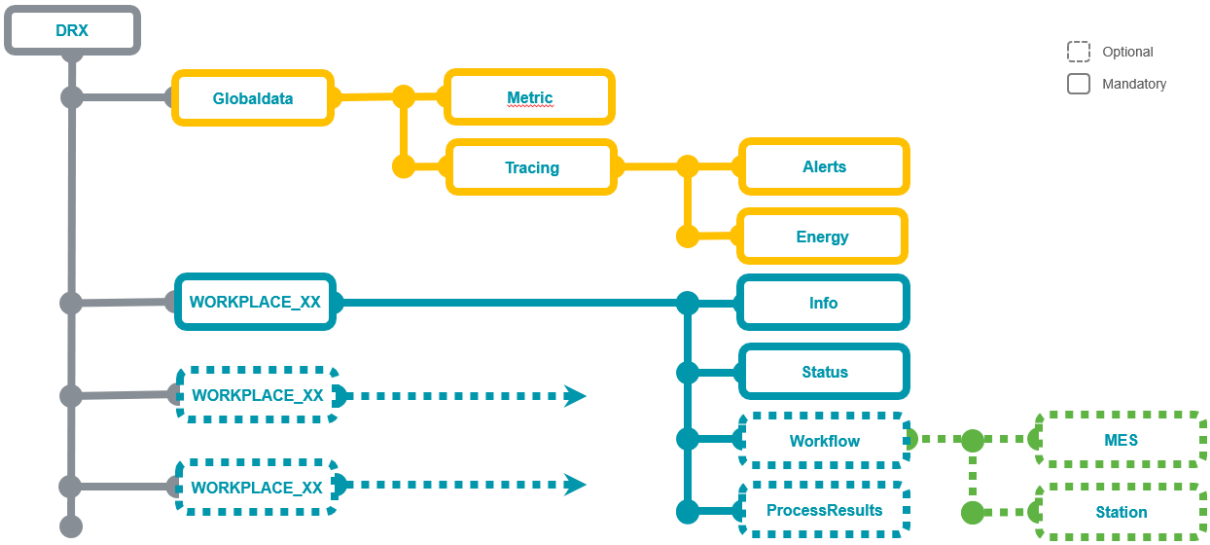


Figure 2: OPC UA Variable Tree

- **Globaldata**
  In this section the assembly line provides typical trace values (power consumption, total time in maintenance, total produced parts, …). These values have to be kept up to date at any time.

- **Workplace**
  Contains all information of a certain Workplace in the station. Simple machines may only have one workplace whereas complex assembly lines can have multiple.

  - **Info**
    The info section contains some static information about the specific workplace. The content will be provided by DRÄXLMAIER.

  - **Status**
    In this section the assembly line provides typical status values which mirror the current production state of a workplace within the station.

  - **Workflow / WorkflowUnload**
    In this sections all variables for fulfilling the workflow for a certain workplace are provided. It is divided into MES and Station section

    - **MES**
      In this section the DRÄXLMAIER MES will provide all needed information to produce a new order on the assembly line (product ID, amounts, variants, …).

    - **Station**
      In this section the assembly line reports the current status (e.g. result, data available, …) and handles the current orders which are in process.

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- **ProcessResults**
  In this section the process data results of each produced part will be reported back to the DRÄXLMAIER MES. The variables in this section are not fixed and therefore specific to each project. These variables (name and datatype) have to be defined together with DRÄXLMAIER at the start of the project and reported to the DRÄXLMAIER MES responsible department at DRÄXLMAIER as soon as possible. Once defined for a certain machine these values are fixed. Dynamic ProcessResults are not allowed!

Note:

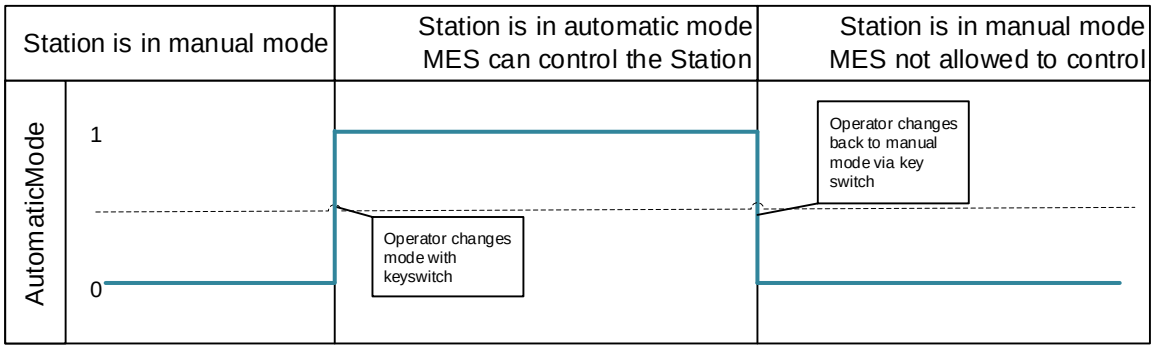
- See the "**OPC UA parameter structure**" at the end of the document for an example of the OPC UA variables structure.
- **Attention:** The amount of process data transferred as via OPC UA should not exceed 3000 variables (this can vary depending on the used OPC UA server and the required cycle time; less cycle time also means less data can be transferred in time).
- If more data, graphs or files need to be transferred other interface techniques, like DATA2CLOUD would be advantageous. In such a case please contact DRÄXLMAIER IT.

4.3 Status section

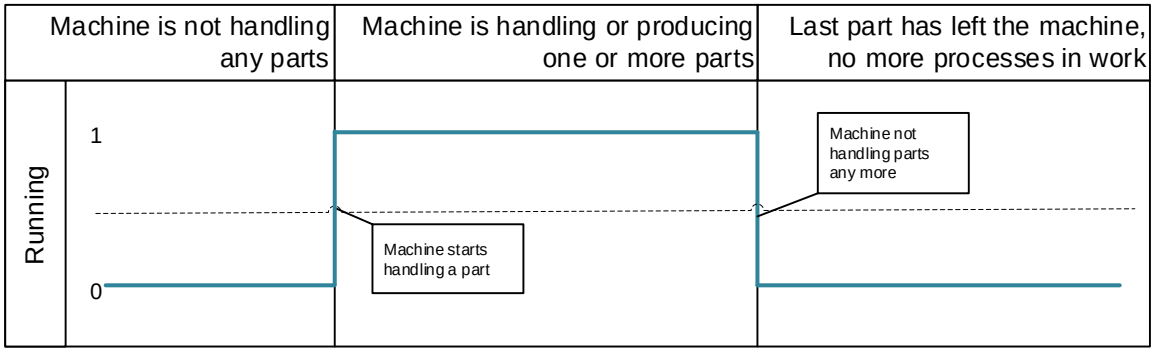
The Status section is supposed to always show the current status of the machine. If a machine consists of more than one station, these variables only represent the corresponding station.
The status variables are for example used for OEE and TEE calculations.

4.3.1 AutomaticMode

When AutomaticMode is switched off, MES will book all running processes to CANCEL, and will reset its variables to initial state. The machine is expected to also be completely initialized when AutomaticMode is switched back on.



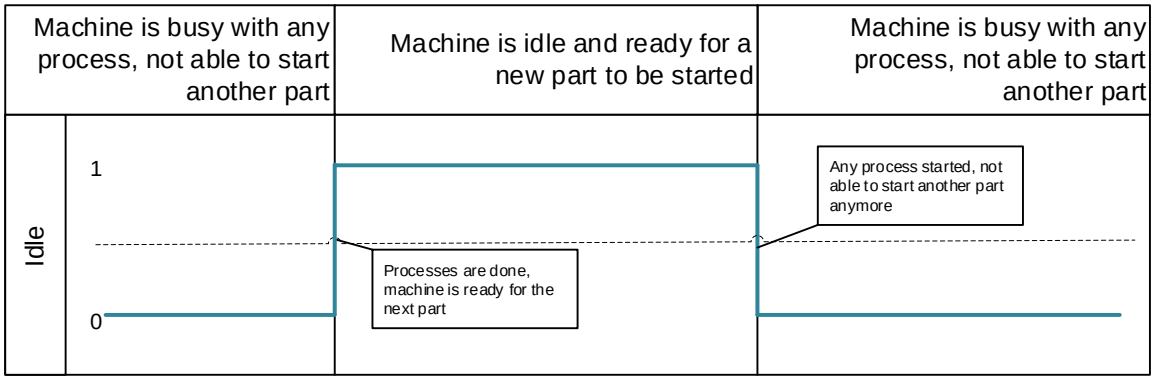
4.3.2 Running



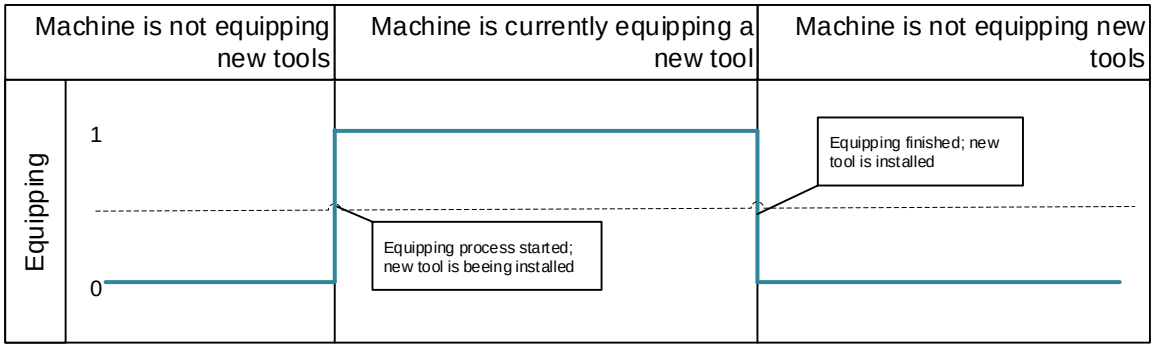
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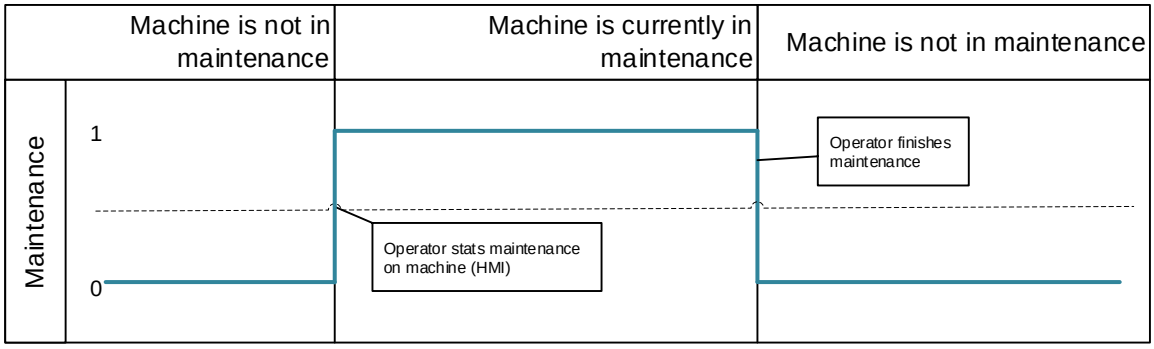
4.3.3 Idle



4.3.4 Equipping

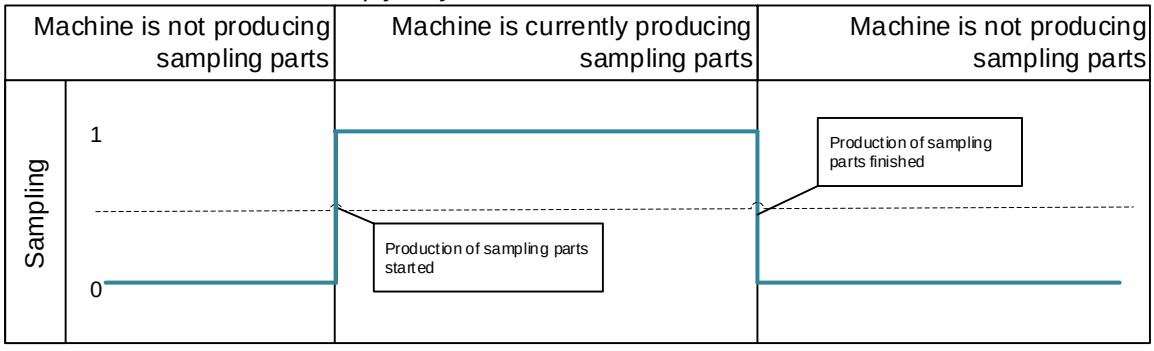


4.3.5 Maintenance



4.3.6 Sampling

Sampling defines the process when one or more parts are run through the machine in order to test or finalize the configuration (teaching / worker self-inspection). If there is no such process requested for the machine the status will simply stay false.

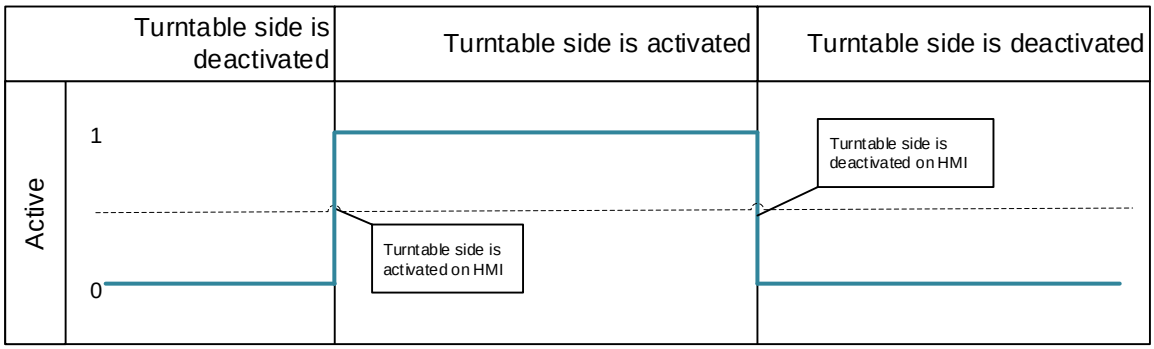


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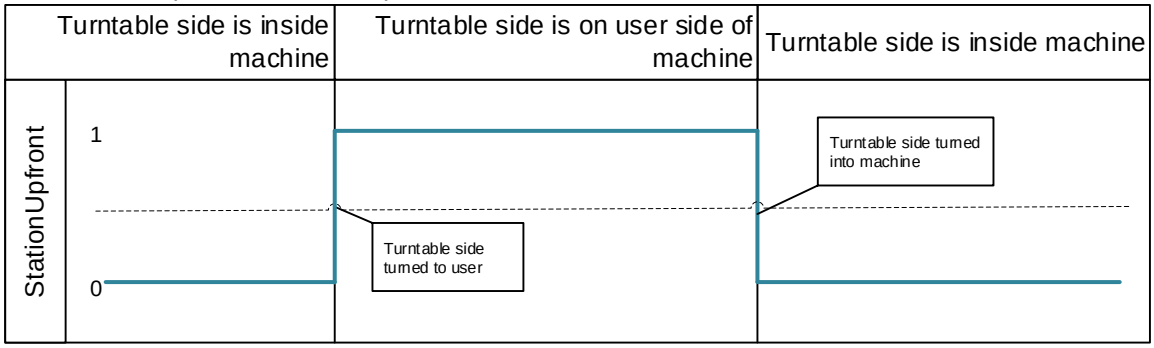
4.3.7 Active

Only turntables. Indicates if the specific side of the turntable is currently active or has been deactivated.



4.3.8 StationUpfront

Only turntables. Indicates if the specific turntable side is currently in the front (on the operators side), or in the back (machine/tool side).



4.3.9 Part counters

The part counters TotalOKParts, TotalNOKParts & TotalScrapParts in the status section should represent the output of a certain station / workplace in the assembly line / machine. If a part leaves a station the counter of that station should be increased according to the production result of the product on that specific station (OK, NOK, SCRAP).
The values will be used for output visualization purposes of single stations inside large machines.

4.4 OPC UA Methods

Not in use

4.5 Tracing data

All data inside the Globaldata/Tracing structure is supposed to show the overall status of the machine. That consists of several counters like TotalOKParts, TotalScrapParts, TotalNOKParts. Those counters shall be increase live during runtime. When they are reset on the HMI or via code the ResetDate shall be filled accordingly to keep the history valid.

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4.5.1 Alerts

The Flag AlertActive shows if there is any active alert on the station (global level). To be able to track the alert down to its root there is an Alerts structure which contains an array of all available functional groups. The array must have a fix size. The Name of the functional Group must also contain a static value. The functional groups themselves contain an AlertActive flag again, which shall indicate if there is an active alert in this specific area. If the flag is true the AlertDetails must be filled accordingly. All alerts which are shown on the HMI of the machine should also be available in this structure (Siemens for example provides a function GetAlarms which can be used to get all alerts)

The functional groups best should be created to match the workplaces on the machine.
E.g.:

- Group one: Name="[MachineID]"; general alerts affecting the whole machine
- Group 2-n: Name="[StationID]"; alerts of workplace 1 - n-1

If a differentiation is not possible at all, only one element in FunctionalGroups which contains all alerts is sufficient.

Each single alert contains three fields: AlertCode, AlertMessage and AlertLevel. AlertCode should be a unique code to identify the alert. AlertMessage contains a string with the important information to the alert. For AlertLevel we propose a differentiation into 3 different levels:

1. ERROR (AlertLevel = 1)

   ERROR describes a severe problem which leads to a immediate stop of the affected workplace/station.
   Signal lights on the machine show red light.
   Examples: Emergency button pressed; broken tool; broken fuses…

2. WARNING (AlertLevel = 2)

   WARNING describes a problem which can be solved by the operator. The signal light is yellow.
   For example: Missing Material

3. INFO (AlertLevel = 3)

   INFO describes a certain situation which is not actually a problem. A Information does not need action of operating personnel but is displayed on the HMI. The signal light is off.
   Examples: Log messages

4.5.2 Energy

This sections should contain all energy monitoring data, dependent on the sensors built in in the machine. E.g. energy consumption, air/gas consumption or environmental KPIs like temperature and humidity. The data in this section should be live. It will be collected in fixed intervals and not within the productflow.

4.6 Process control flags

4.6.1 PartAvailable (optional)

When a fixture and part presence sensors are available on the machine this flag is used to signalize that a part has been inserted and locked in the station. In such cases the flag may be used to start the process on the MES display.
The flag needs to stay active as long as the part is available in the fixture. If the part cannot be locked mechanically and someone takes it out before the process is completed this might cause a CANCEL booking on MES side if the release was already given before.

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4.6.2 ProductIDAvailable (optional)

This flag is an extension to the PartAvailable flag, when the machine is equipped with automatic scanners. It shall be set active, when

- Part is available on station
- AND Barcode/RFID is scanned automatically and filled into ProductID

This flag only indicates the successful scan of the main barcode. If the machine is equipped with more scanners, the additional scans will be handled separately in specific actions.
When used, the flag needs to stay active as along as the part is available in the fixture. If the part can be locked mechanically and someone takes it out before the process is completed this might cause a CANCEL booking on MES side.

4.6.3 MES_Release

MES_Release signalizes that the machine can start with production. Before it is set active MES needs to ensure that all production related information is available and up to date (this includes MES_ProductID, MES_Action, and all variables which are needed to define the job, including MES_ProcessParameters).
MES_Release stays active during the performed action. If more than one action is performed at a station it will be set to false during the handshake and set true again with the next action.
If MES_Release is set to false during a process it indicates that there was an unexpected error on MES side. In such a case MES might not react properly to the handshake at process end. In such a case the machine has to abort its actions as soon as possible.

4.6.4 DataAvailable

DataAvailable signalizes the end of a production process. Before this flag is set active, the machine needs to ensure that:

- ProductID and Result are filled according to the outcome of the process
- All process results of the performed action are up to date

4.6.5 MES_DataProcessed

After the machine has ended the process and provided the process results, MES reads and saves all required data. To signalize that the data on the machine can be set back to its initial state MES_DataProcessed is set to true (in the same step MES_Release and MES_Action will be set back). MES_DataProcessed will remain true until DataAvailable and Result are initialized again.
After this handshake the machine shall be ready to either accept the next action triggered by MES or release the part and wait for the next one (e.g. when only one specific action is planned for a machine)

4.6.6 MES_CancelProcess

When a part has been inserted and locked or scanned, but an error occurs on MES side MES has to option to cancel the process via MES_CancelProcess. When this flag is activated MES expects the machine to release the part and set the associated variables back to their initial state.
When an action is already in progress MES it not supposed to attempt cancelling the process. When necessary this shall be done on machine side (e.g. on the HMI or with a key switch). When cancellation is done on machine side the handshakes must be performed in the standard way (Result will be 3 in this case)

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4.7 Workstep and variant indicators

4.7.1 MES_Action

Is a string variable to define the process which shall be started. The actions which need to be available on a certain machine need to be defined in advance.
Examples of common actions:
SCORE
SCREW (can be SCREW_1 if more than one screwing steps are available)
WELD
PUNCH
CLIP
CHECK
LAMINATION
FOAM
BACKFOAM
SCAN
SEW
UNLOCK
LOCK
MOUNT
COMMISSION
SCRAP
RELEASEPART
TEACH
SAMPLE
START
CANCEL
INTERRUPT
REORDER

4.7.2 MES_ProductID

Currently processed order, FP-Number or article. Used to identify a certain product and link it to its process data. When ProductID is filled by the machine with an automatic scanner MES will fill this variable with the same ID. When the barcode is scanned by the user, or generated in the system MES will update this field accordingly and machine shall copy the info to its info when a process is started (on MES_Release)

4.7.3 MES_ProcessParameters

MES_ProcessParameters is a structure which shall contain all data need by the machine as input to start production. This may include variant information, material lists or other process parameters. As this data is highly specific for a certain machine, the content of the structure needs to be defined separately for each machine.

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5 Workflow and handshakes

5.1 Standard Handshake

This workflow describes a handshake which is defined as the standard for each MES_Action. It will be always the same flow, independent from the chosen action and the base for communication between the assemly line and MES.

Before an action is started MES will store all data required for production on the OPC server. This includes the MES_ProductID and MES_ProductType as identifier for the part which shall be produced and MES_PartsToProduce to specify how many parts of this type need to be done.
Depending on the product and machine additional information can be required, this data will be provided in the structure MES_ProductParameter. This can include variants, all kinds of product specifications or material lists.

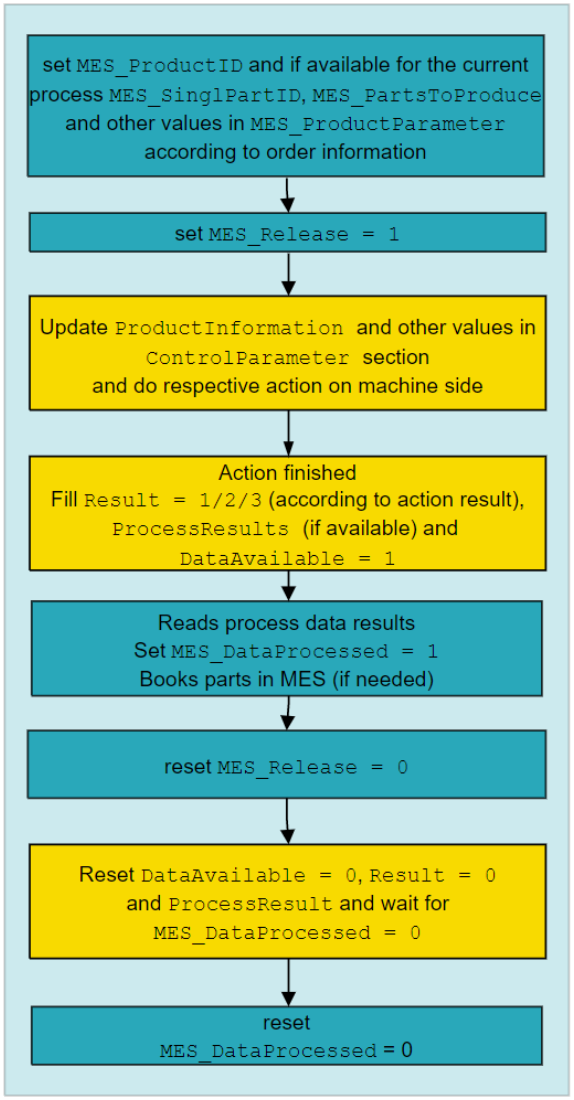


Figure 3: Standard Handshake

For the detailed workflows available on the machine please check the detail descriptions.

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5.2 Standard workflow (flow chart)

5.2.1 Manual Scan to start order

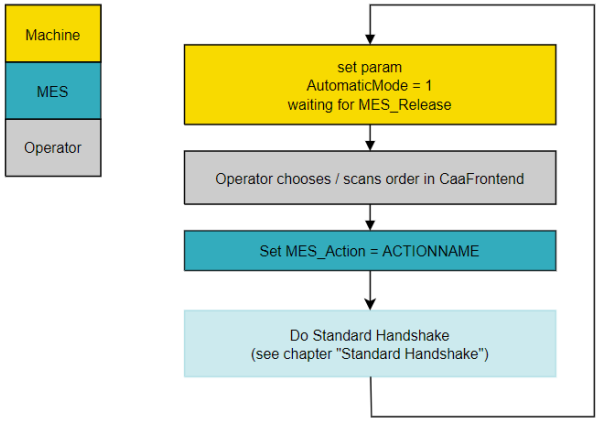


Figure 4: Manual Scan

5.2.2 Auto Scan to start order

This workflow describes the process if a specific label has to be scanned for starting the order. This can be done either with an Automatic scanner or by scanning by user.

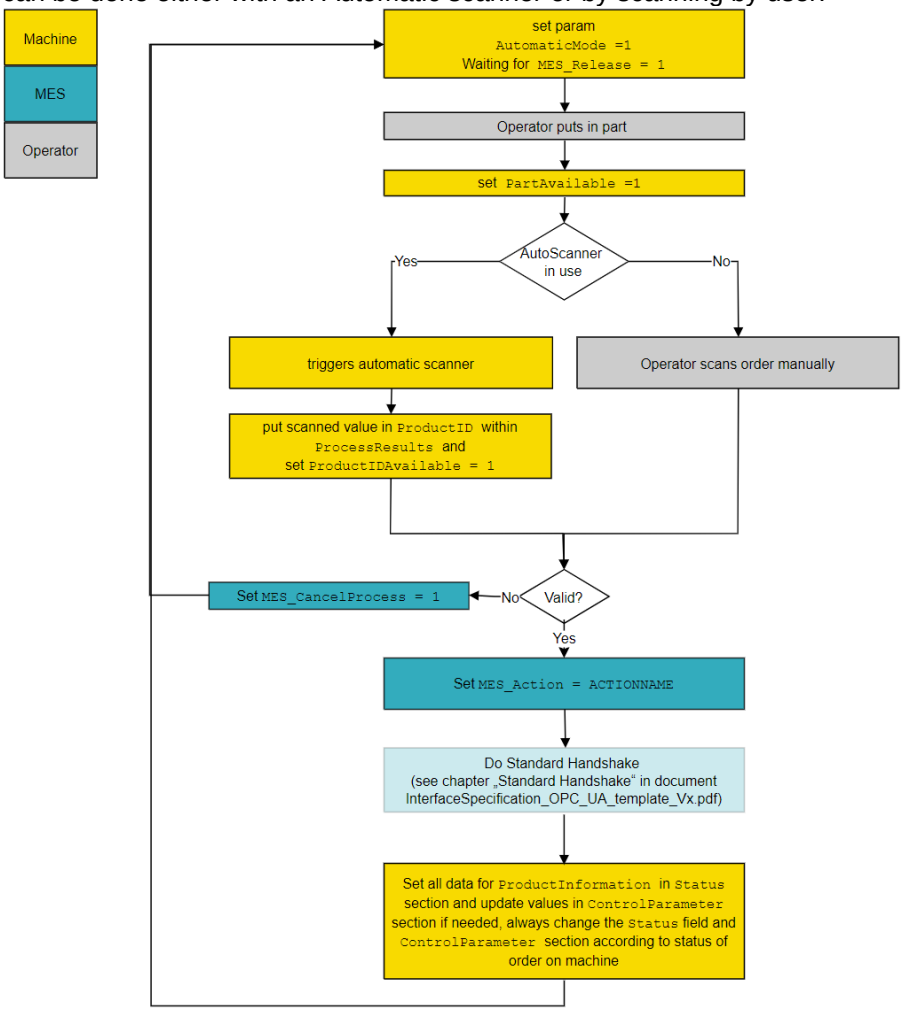


Figure 5: Standard workflow with automatic scanner for ProductID

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5.2.3 Auto scan inside process

The handshake for an integrated scanner that will be used inbetween the standard handshake (e.g. for a commissioned part) will be the following:

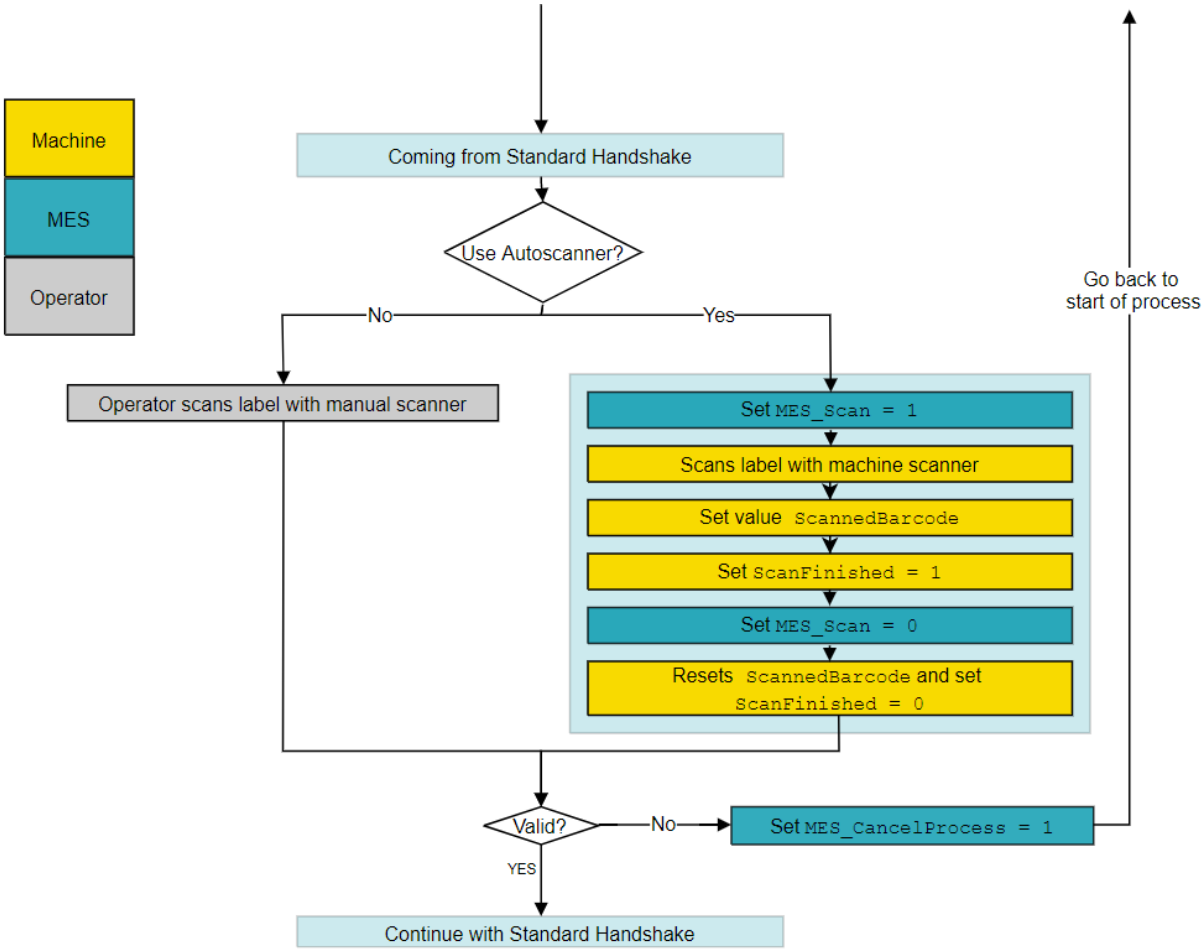


Figure 6: Standard workflow with automatic scanner inside process

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This process will be called from inside the StandardHandshake. Based on this the complete process will look be the following:

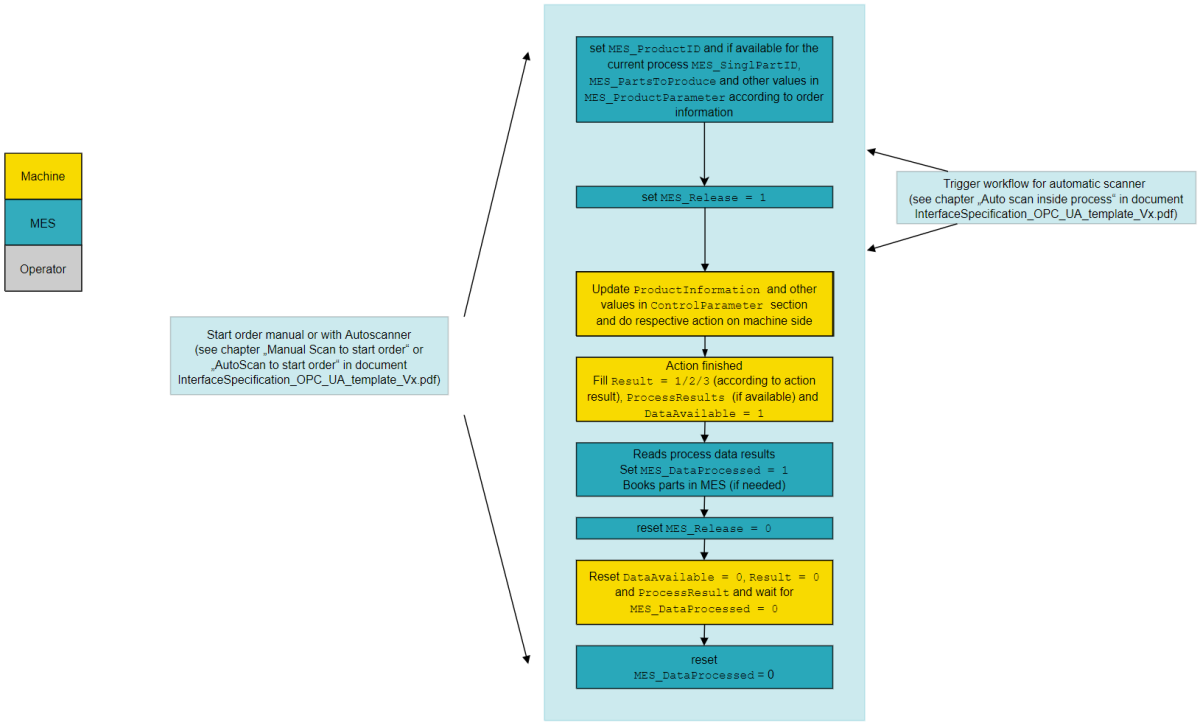


Figure 7: Standard workflow with automatic scanner inside process

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5.3 Standard workflow (swim lane)

Single actions can be skipped in certain cases. The station must be able to perform any action when the correct set of input data is given.

5.3.1 Manual Scan to start order

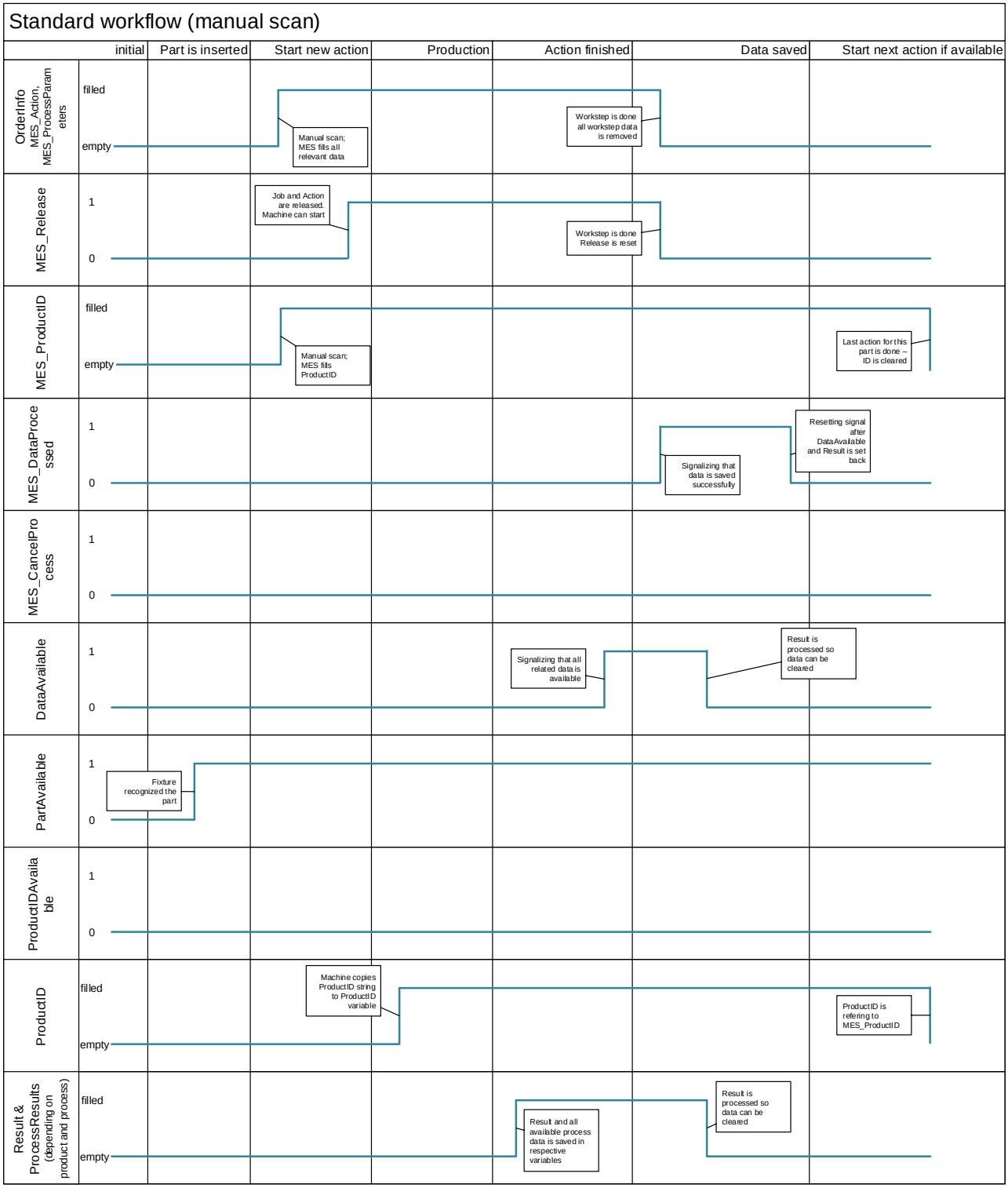


Figure 8: Swimlane of standard workflow (manual scan)

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5.3.2 Auto Scan to start order

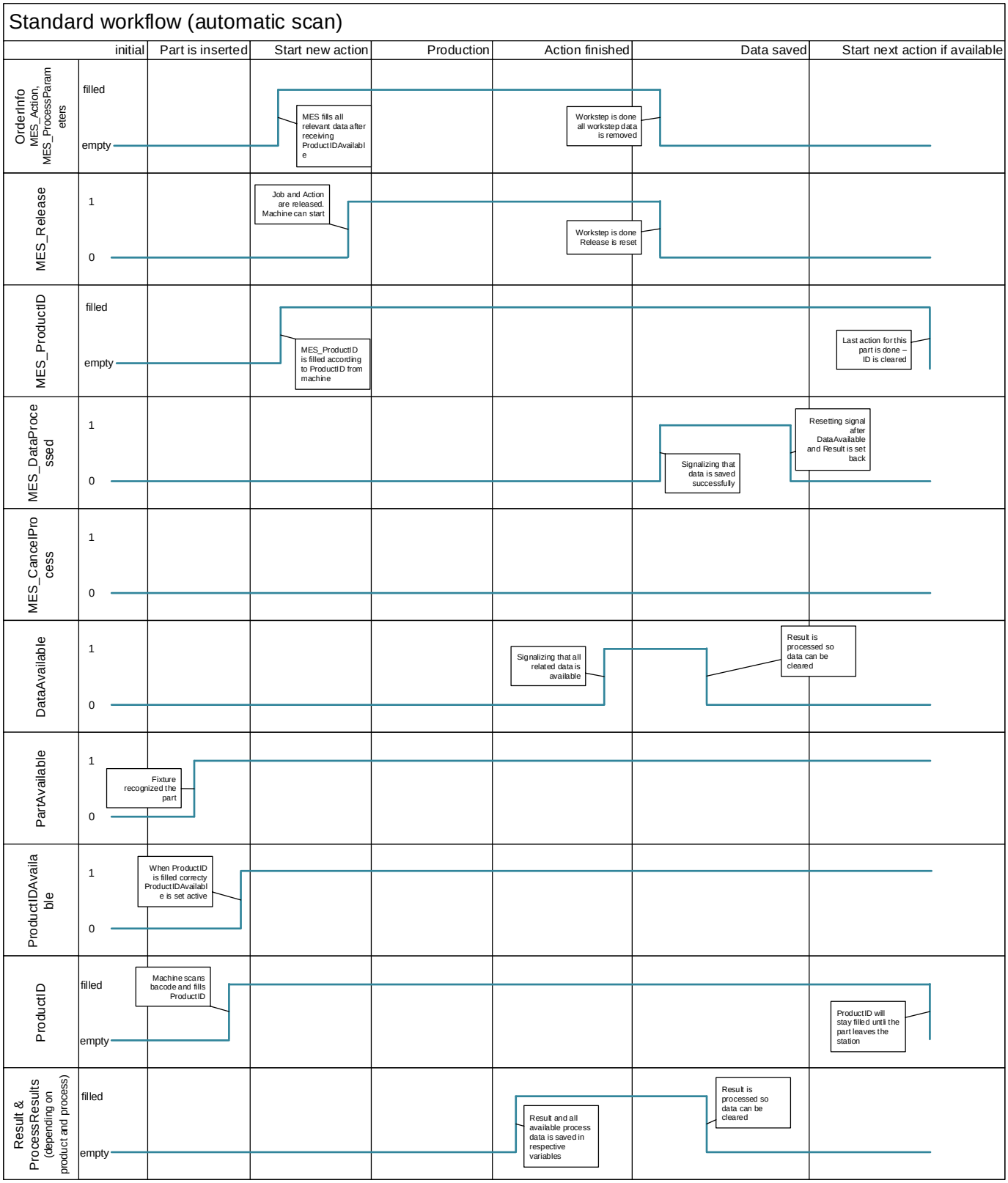


Figure 9: Swimlane of standard workflow (automatic scan)

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5.4 Cancellation scenarios

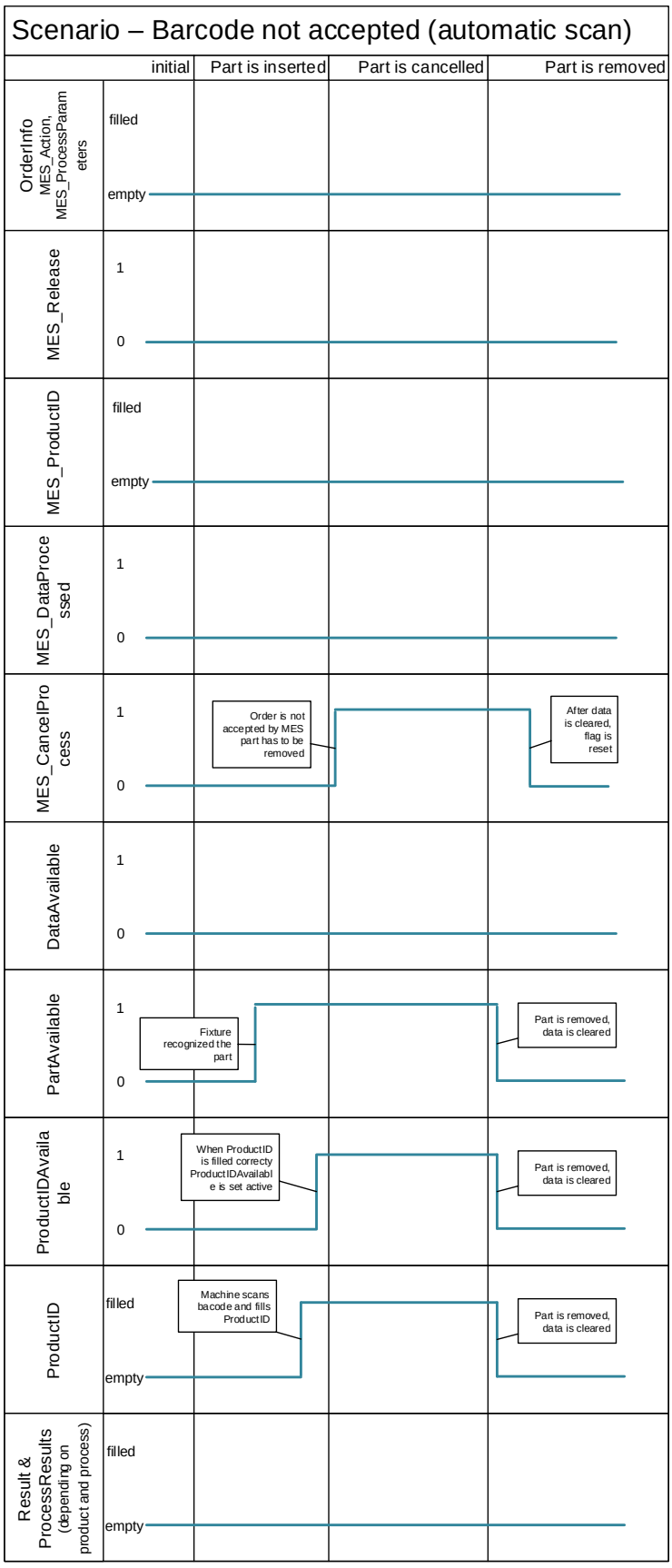


Figure 10: Swimlane of cancel scenario – Barcode not accepted

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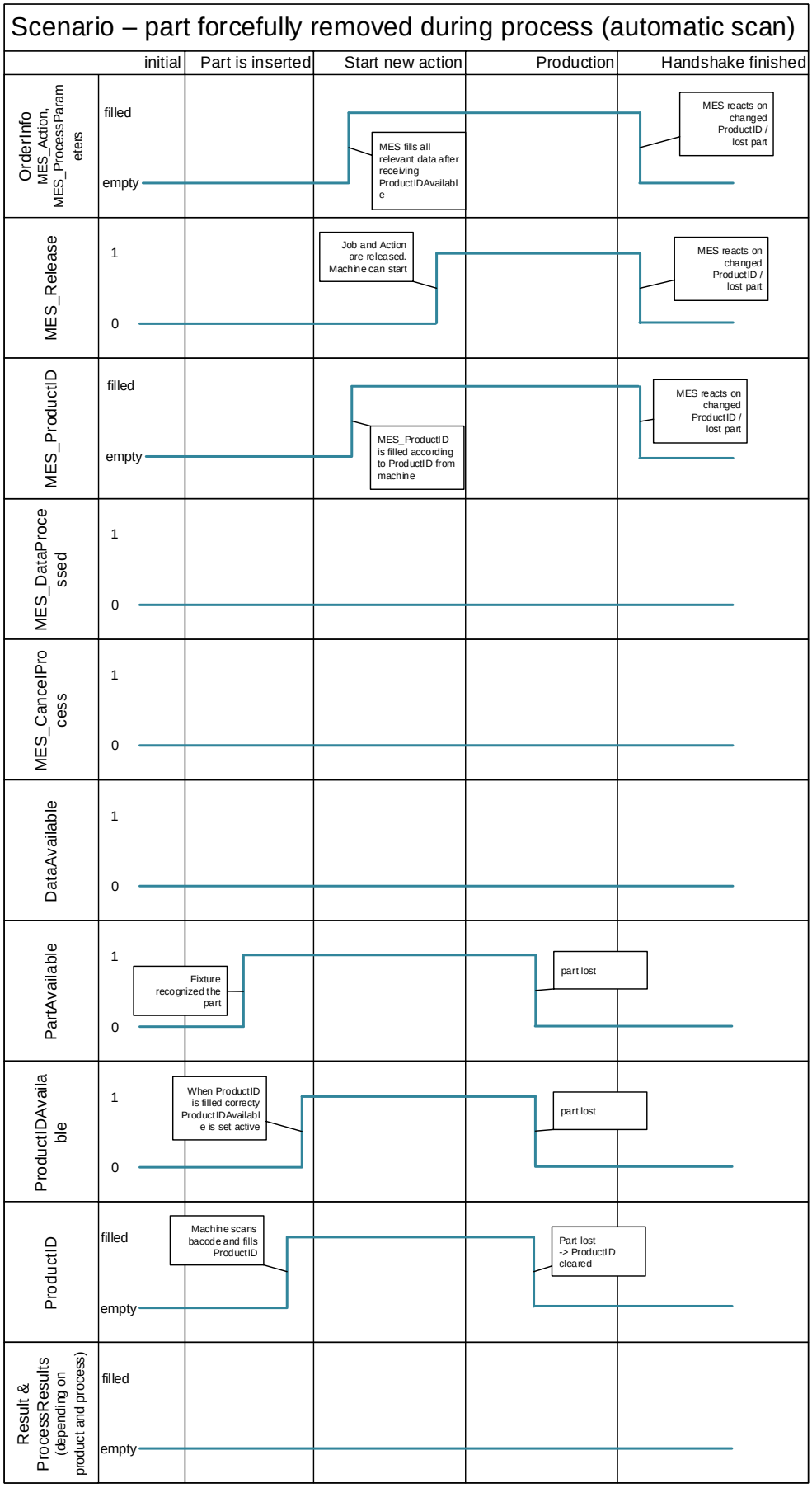


Figure 11: Swimlane of cancel scenario – Part forcefully removed

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6 MES Input parameters

All the parameters needed to transfer data from DRÄXLMAIER MES to the station are located in the structure Workflow > MES. The mandatory subset of values are available on top level in that structure. For specific requirements in certain cases the substructure MES_ProductParameters can be used. Examlpes for such specific variables are e.g. steering type, job numbers, material lists…

Assignments HOST – SAP – OPCUA wordings

		Today		synapsis	
	Naming on OPCUA Server	Format	Example	Format	Example
Production order ID	MES_ProductID	Consists of Productnumber + Batch Number + Single Part Number (optional)	RO123AS115ST45AB 0001 RO123AS115ST45AB	Consists of Productnumber + Batch Number + Single Part Number (optional)	123456789-001P123456789000001 123456789-001P123456789
FP Nr.	MES_ProductType	10 digits	RO123AS115	13-digits DRÄXLMAIER Part number (Part number + „-“ + changelevel) ZMAT	123456789-001
Material number	MES_ProductType	8 digits for resin	K0145632		
SAP part numbers	MES_ProductType	12 digits	123456789/01		
Equipment main number	EquipmentMainNumber	5 digits	87245	13-digits DRÄXLMAIER Materialnumber ZFHM	500000023-001
Equipment exemplar number	EquipmentExemplarNumber	3 digits (added to Equipment main number)	001	18 digits – only 10 digits used (unique with assign-ment to ZFHM Materialnr)	1000000034
Workcenter	MES_Workcenter	6 digits alphanumeric	081520	2 digits alphanumeric + 5 digits process number	IP30405

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7 Process results and return values

Whereas the overall result of a workflow is located in structure Workflow > Station, more detailed information on the production/test result is to be added in the structure ProcessResults. The data is split into following subsets.

7.1 GeneralData

GeneralData is a mandatory data set containing are more detailed overview on the job performed at the station. The contained fields need to be filled according to the stations output.

- ProductID:
  - ID of the processed part, order or series. This field links to the variable MES_ProductID and has to be filled for each result as on some machines multiple orders can be active at the same time.
- ProductType:
  - Type of the currently processed part, order or series. Has to be filled in the same logic as ProductID.
- PartsNOK, PartsOK, PartsScrap
  - Has to be filled according to the output of the station. If the machine has single part responses one of the fields is filled with 1 and the other two with 0. In case a whole series is produced the counters have to be filled according to the single results.
- ErrorCodesNOK, ErrorCodesScrap, ErrorMessagesNOK, ErrorMessagesScrap
  - In case a part is NOK or Scrap an according error code and message has to be provided and filled with the reason why the result was not good!
    - NOK definition: Process was not OK but part can possibly be reworked
    - Scrap definition: Process was not OK and there is no chance for rework (e.g.if machine contains a NOK part cutter)
- TotalProcessingTime
  - Time in minutes of the whole production process on the station.
  - If more than one part is produced in a job (batch production) the time should be measured from the first part of the batch to the last one.
  - Detailed measuring frame has to be defined for each machine between the manufacturer and the ordering department (DRÄXLMAIER Industrial Engineering)

    The reason is synchronization of the calculation on machine side with the calculation on BOP side.

7.2 MaterialConsumption

Material consumption considers all materials which are consumed and scanned automatically by the machine, e.g. parts which are taken out of a warehouse by a robot and assembled to the product, screws or clips which are used through a feeder… For these materials several values have to be reported (see parameter structure).
In general the consumption is split into two structures, regular and additional consumption. Regular consumption means everything which should be consumed according to the bill of material (BOM) and leaves the station on the intended position. Material which exceeds the BOM, or was not mounted as it was supposed, has to be filled in the structure AdditionalConsumption.
Every material which is supposed to be consumed according to BOM needs to be present in the structure, if the process is aborted before the material is needed the consumption is to be set to 0

Example:

- Clip was mounted and leaves the station in the intended position on the part
  - ➔ RegularConsumption
- Clip was lost during handling, position unknown
  - ➔ AdditionalConsumption
- Clip could not be mounted and was thrown in the scrap bin
  - ➔ AdditionalConsumption
    - Clip that successfully replaces the previous one -> Regular Consumption
- Process aborted before Clip was needed

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➔ RegularConsumption -> consumption = 0

All materials which have already been commissioned on previous stations are excluded from this logic.

7.3 ProcessTimes

The ProcessTimes array should contain start and end times of all relevant processes and transfers in the station for further evaluation. The different entries can overlap each other. Examples: total process, carrier entry, screwing, cleaning, carrier exit…
At a very minimum this array has to contain an entry of the total process with start and end time stamps.

Example (Json formatting for better readability)


```
{
    "ProcessTimes": [
        {
            "EquipmentID": "004001FX_06",
            "OperationID": "004001FX_06_cleaning",
            "Description": "Batterierahmen reinigen",
            "StartTime": "2022-04-11T11:03:50.3833455Z",
            "EndTime": "2022-04-11T11:05:04.8359223Z"
        },
        {
            "EquipmentID": "002001FX_10",
            "OperationID": "002001FX_10_transfer",
            "Description": "R10 Holen Batterierahmen Reinigung Ablegen E-Cart",
            "StartTime": "2022-04-11T11:05:04.8359182Z",
            "EndTime": "2022-04-11T11:10:26.8427106Z"
        },
        {
            "EquipmentID": "002001HR_10",
            "OperationID": "002001HR_10_transfer",
            "Description": "R10 Deckel GLT auflegen",
            "StartTime": "2022-04-11T11:10:26.8999961Z",
            "EndTime": "2022-04-11T11:10:58.0064457Z"
        },
        {
            "EquipmentID": "E-Cart",
            "OperationID": "E-Cart_exit",
            "Description": "E-Cart Wechselzeit Station 020",
            "StartTime": "2022-04-11T11:10:58.4002381Z",
            "EndTime": "2022-04-11T11:17:54.8710314Z"
        },
        {
            "EquipmentID": "002001",
            "OperationID": "002001_runtime",
            "Description": "Durchlaufzeit Station 020",
            "StartTime": "2022-04-11T11:03:50.0497011Z",
            "EndTime": "2022-04-11T11:10:58.0250052Z"
        }
    ]
}
```


7.4 Process specific data

Additional to the predefined structures ProcessResults contain specific results for each process performed on the station. These values are defined separately and will be added to the parameter structure during the project phase.

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8 Glossar

HMI	Human-Machine Interface
M2M Communication	Machine-to-machine communication
OEE	Overall equipment effectiveness
OPC UA	Open Platform Communications Unified Architecture
TEE	Total effective equipment (productivity)
Shopfloor Visualization	A DRX internal software for evaluating machine data and processes

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